

Capstone Project Online Retail Customer Segmentation

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1. Problem Statement:

In today's competitive e-commerce environment, understanding customer purchasing behavior is essential for improving business performance, enhancing customer satisfaction, and maximizing revenue. The online retail store possesses a large transactional dataset (online_retail.csv) containing 387,961 records, including details such as invoices, products, quantities, prices, customer IDs, and countries.

However, the raw data alone does not provide clear insights into customer behavior patterns. The company currently lacks a structured approach to identify key customer segments, analyze purchasing trends, and derive actionable insights for targeted marketing and strategic decision-making.

Therefore, the problem is to analyze the historical transaction data to uncover meaningful patterns in customer purchasing behavior and segment customers into distinct groups based on their buying habits.

This analysis aims to help the business identify high-value and loyal customers, detect customers who are at risk of churn, understand purchasing frequency and spending behavior, and enable data-driven marketing and personalization strategies.

By solving this problem, the company can improve customer retention, optimize marketing efforts, and enhance overall business growth.

2. Problem Objective:

The main objectives of this project are as follows:

- To analyze the customer transaction data of an online retail store to understand purchasing behavior.
- To identify patterns in customer activity such as purchase frequency, recency, and spending habits.
- To perform customer segmentation based on behavioral metrics.
- To apply RFM (Recency, Frequency, Monetary) analysis for evaluating customer value. •

To implement clustering techniques (such as K-Means) to group customers into meaningful segments.

- To identify high-value, loyal, and at-risk customers.
- To understand variations in purchasing trends across different customer groups.
- To provide data-driven insights for improving marketing strategies .
- To enhance customer retention and increase customer lifetime value .
- To support better business decision-making using analytical results.

3. Data Description

The dataset used in this project is **online_retail.csv**, which contains transactional data of an online retail store. This dataset provides detailed information about customer purchases and is used to analyze customer behavior and perform segmentation.

Dataset Overview

- **Total Records:** 387,961
- **Total Features:** 8
- **Data Type:** Transactional data
- **Nature:** Real-world retail purchase data

Feature Description

- **Invoice:** Unique identification number for each transaction
- **StockCode:** Unique code assigned to each product
- **Description:** Name or description of the product
- **Quantity:** Number of units purchased per transaction
- **InvoiceDate:** Date and time when the transaction occurred
- **Price:** Price per unit of the product
- **CustomerID:** Unique identifier for each customer
- **Country:** Country where the customer is located

Key Characteristics of the Data

- The dataset contains **multiple transactions per customer**, enabling behavioral analysis
- Each invoice may include **multiple products**, representing a single purchase session
- The data includes **time-based information**, which is useful for analyzing purchase trends
- Some records may contain **missing values**, especially in the CustomerID field
- Negative or zero values in Quantity may indicate **product returns or cancellations**

Derived Features (Used in Analysis)

To perform customer segmentation, additional features are created:

- **Total Amount:** Calculated as $Quantity \times Price$
- **Recency:** Number of days since the customer's last purchase
- **Frequency:** Total number of transactions made by a customer
- **Monetary Value:** Total amount spent by the customer

4. Data Pre-processing Steps and Inspiration

The data preprocessing stage is a crucial part of the project, as it ensures that the dataset is clean, consistent, and suitable for analysis and modeling. The following steps were performed during data preprocessing:

- Removed missing values, especially in the **CustomerID** field, as customer identification is essential for segmentation
- Filtered out records with negative or zero quantities to eliminate cancelled transactions and returns
- Checked and removed duplicate records to maintain data integrity
- Converted the **InvoiceDate** column into proper datetime format for time-based analysis
- Created a new feature **Total Amount** by multiplying Quantity and Price
- Aggregated transaction-level data into customer-level data for analysis
- Calculated RFM (Recency, Frequency, Monetary) metrics:
 - o **Recency:** Number of days since the last purchase
 - o **Frequency:** Total number of transactions per customer
 - o **Monetary:** Total amount spent by the customer
- Handled outliers in Monetary and Frequency values to reduce skewness
- Applied feature scaling (such as normalization or standardization) to bring all variables to a comparable scale
- Prepared the final dataset for clustering by selecting relevant features (RFM values)

The preprocessing steps were inspired by standard data cleaning practices and the need to transform raw transactional data into meaningful customer-level insights suitable for segmentation and analysis.

5. Choosing the Algorithm for the Project

In this project, the primary objective is to segment customers based on their purchasing behavior. Since there are no predefined labels or categories available in the dataset, an **unsupervised learning approach** is most suitable.

The algorithm chosen for this project is the **K-Means Clustering Algorithm**.

K-Means is a widely used clustering technique that groups similar data points together based on their characteristics. It works by dividing the dataset into a predefined number of clusters (K), where each data point belongs to the cluster with the nearest mean (centroid).

The algorithm was applied on the RFM (Recency, Frequency, Monetary) features, which represent customer behavior in terms of how recently they purchased, how often they purchase, and how much they spend.

K-Means clustering helps in:

- Grouping customers with similar purchasing patterns
- Identifying distinct customer segments such as high-value, regular, and low-value customers
- Simplifying complex data into meaningful clusters for business understanding

The selection of K-Means is appropriate for this project due to its simplicity, efficiency, and effectiveness in handling large datasets like the online retail data.

By applying this algorithm, the project is able to segment customers into meaningful groups, which can then be used for targeted marketing and business decision-making.

6. Motivation and Reasons for Choosing the Algorithm

The selection of the K-Means clustering algorithm for this project is motivated by several practical and analytical considerations.

- The dataset does not contain predefined labels, making **unsupervised learning** the appropriate approach
- K-Means is simple to understand and easy to implement, making it suitable for practical applications
- It is computationally efficient and performs well on large datasets like the online retail dataset
- The algorithm effectively groups customers based on similarity in their purchasing behavior
- It works well with numerical features such as Recency, Frequency, and Monetary values •

K-Means provides clear and interpretable clusters, which are useful for business decision-making

- It allows flexibility in choosing the number of clusters (K) based on evaluation methods such as the Elbow Method
- The algorithm helps in identifying distinct customer segments such as high-value, regular, and low-value customers
- It supports data-driven marketing strategies by enabling targeted customer segmentation

Overall, K-Means clustering is a suitable choice for this project as it efficiently transforms complex customer data into meaningful and actionable segments.

7. Assumptions

The following assumptions were made during the development of this project to simplify the analysis and modeling process:

- Customers with missing **CustomerID** values are excluded, assuming they cannot be reliably identified for segmentation
- Negative or zero **Quantity** values represent product returns or cancellations and are removed from the analysis
- The **Price** of products is assumed to be accurate and consistent across transactions
- Each **Invoice** represents a valid transaction made by a customer
- Customer behavior can be effectively represented using RFM (Recency, Frequency, Monetary) metrics
- The dataset is assumed to be representative of overall customer purchasing behavior
- Outliers in spending and frequency are treated or limited to avoid distortion in clustering results
- The number of clusters (**K**) chosen for K-Means is assumed to be optimal based on evaluation techniques (e.g., Elbow Method)
- Customers within the same cluster are assumed to have similar purchasing behavior
- Time-based analysis assumes that the dataset covers a sufficient time period to reflect realistic customer activity

These assumptions help in building a structured and efficient model while ensuring meaningful interpretation of the results.

8. Model Evaluation and Techniques

To evaluate the performance and effectiveness of the customer segmentation model, several techniques and analytical methods were used.

- **Elbow Method:**
This technique was used to determine the optimal number of clusters (K) for the K-Means algorithm. It works by plotting the Within-Cluster Sum of Squares (WCSS) against different values of K and identifying the point where the rate of decrease sharply changes (the “elbow point”).
- **Silhouette Score (if applied):**
This metric was used to measure how well each data point fits within its assigned cluster. A higher silhouette score indicates better-defined and well-separated clusters.
- **Cluster Visualization:**
Graphical representations such as scatter plots were used to visualize the distribution of customers across clusters based on RFM features. This helps in understanding the separation and compactness of clusters.
- **Feature Scaling Validation:**
Proper scaling (normalization/standardization) was verified to ensure that no single feature (Recency, Frequency, Monetary) dominates the clustering process.
- **Cluster Profiling:**
Each cluster was analyzed by calculating average RFM values to interpret the characteristics of different customer groups.

Evaluation Results and Observations

- The Elbow Method helped in selecting an optimal number of clusters that balances accuracy and simplicity.
- The clusters formed were distinct and meaningful in terms of customer behavior.
- High-value customers showed high monetary and frequency values with low recency.
- Low-value or at-risk customers showed low frequency and monetary values with high recency.
- The segmentation results were interpretable and aligned with real-world business scenarios.

Overall, the evaluation confirms that the K-Means clustering model effectively segments customers into meaningful groups, enabling actionable business insights.

9. Inferences from the Project

The analysis and customer segmentation performed in this project provided several meaningful insights into customer purchasing behavior.

- Customers can be clearly divided into distinct segments based on their Recency, Frequency, and Monetary values
- A group of **high-value customers** was identified, characterized by frequent purchases, recent activity, and high spending
- A segment of **regular customers** was observed, who purchase consistently but with moderate spending
- **At-risk customers** were identified as those who have not made recent purchases and show low engagement
- A group of **low-value or occasional customers** was found, with low purchase frequency and minimal spending
- The majority of revenue is contributed by a small percentage of high-value customers
- Customers with high recency (long time since last purchase) indicate potential churn risk
- Frequency and monetary values are strong indicators of customer loyalty and importance

These insights can help the business. Customer segmentation enables better understanding of purchasing trends across different groups

- Design targeted marketing campaigns for different customer segments
- Focus on retaining high-value customers
- Re-engage at-risk customers through offers and promotions
- Improve overall customer satisfaction and profitability

Overall, the project successfully transforms raw transactional data into actionable insights that support strategic business decisions.

10. Future Possibilities of the Project

The current project provides a strong foundation for customer behavior analysis and segmentation; however, there are several opportunities to enhance and extend the work in the future.

- Implementation of more advanced clustering algorithms such as **Hierarchical Clustering** or **DBSCAN** for improved segmentation
- Integration of additional customer data such as demographics, browsing behavior, or feedback for deeper analysis
- Development of a **real-time recommendation system** based on customer purchase history
- Use of predictive models to forecast **customer lifetime value (CLV)** and future purchasing behavior
- Application of **churn prediction models** to proactively identify and retain at-risk customers
- Deployment of the model into a **business dashboard (e.g., Power BI or Tableau)** for real-time insights
- Automation of the data pipeline for continuous updating and monitoring of customer segments
- Personalization of marketing campaigns using machine learning techniques
- Expansion of analysis to include **seasonality trends and product-level insights**
- Integration with CRM systems for better customer relationship management

These future enhancements can significantly improve the effectiveness of the system and provide deeper, more actionable insights for business growth and competitive advantage.

11. Conclusion

In this project, customer transaction data from an online retail store was analyzed to understand purchasing behavior and segment customers into meaningful groups. By applying data preprocessing techniques and RFM (Recency, Frequency, Monetary) analysis, the raw transactional data was transformed into a structured format suitable for analysis.

The K-Means clustering algorithm was successfully implemented to group customers based on their purchasing patterns. The resulting segments provided clear insights into different types of customers, such as high-value, regular, and at-risk customers.

The analysis revealed that a small group of customers contributes significantly to the overall revenue, while some customers show signs of inactivity and potential churn. These insights can help the business design targeted marketing strategies, improve customer retention, and optimize overall performance.

Overall, the project demonstrates how data-driven techniques can be effectively used to extract valuable insights from large datasets and support informed business decision-making.

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